

Auditing the OPAS Orbital Proximity Alert System’s Safe-Window Scanner

Nadim Baboun¹

¹Al-Quds University

September 29, 2026

Abstract

The Orbital Proximity Alert System (OPAS) is a web-based collision avoidance tool that estimates safe launch windows from two-line element (TLE) sets and the SGP4 propagation model. This study audits its `/safe-windows` endpoint against a dense, independently validated reference calculation. Every window the original scanner reported as safe in three crowded low Earth orbit shells (ISS, Starlink, and sun-synchronous) was, per the reference, unsafe for its entire duration, while a geostationary transfer control case was correct. The failure is traced to insufficient discrete trajectory sampling relative to object density. A corrected scanner—dense 600-point sampling, an age-scaled 10–15 km screening radius, and one-minute temporal verification—returns one window each for ISS (9.6% false-safe) and Starlink (21.1% false-safe), with all ten residual close approaches traced to the corrected scanner’s finite waypoint spacing, and no window for SSO. Direct enumeration of clear launch-time intervals in the dense reference, independent of the scanner’s own window-forming algorithm, confirms that no interval of at least 15 minutes exists for Starlink or SSO at the corrected radius — so the reported Starlink window is itself false-safe, and SSO’s absence of any window is correct — while showing that the same algorithm can also miss genuinely clear ISS windows that do exist. We compare these choices with documented NASA conjunction-assessment practice and propose anisotropic, continuous swept-volume screening as future work.

1 Introduction

The Orbital Proximity Alert System (OPAS) is a lightweight web tool for assessing collision risk during satellite and rocket launches. Given a launch site, target orbit, and inclination, its `/safe-windows` endpoint scans forward in time and returns time intervals during which, according to the tool, the planned trajectory stays clear of tracked orbital debris and satellites¹. The system ingests two-line element (TLE) sets from Space-Track, propagates object positions using the SGP4 model via the Skyfield library, and screens a simplified circular-orbit vehicle trajectory against a catalog of tracked objects near the target altitude.

OPAS was developed through rapid, AI-assisted prototyping rather than a formal aerospace engineering process. Its core output — a claimed “safe launch window” — is precisely the kind of claim that should not be trusted without independent verification, particularly given that the tool’s internal screening logic relies on coarse temporal and spatial sampling to keep response times practical for a web API. This report presents an empirical audit of that screening logic: whether the windows OPAS reports as safe are, in fact, free of close approaches, and if not, why.

To answer this, we built a dense, independently-implemented reference calculation that estimates the true minimum distance between the launch vehicle and every tracked object at

¹Source code: <https://github.com/nadeemtsf/OPAS>.

one-second resolution, using the same underlying orbital data and propagation model as OPAS itself, so that any disagreement between the two can be attributed to sampling density rather than a difference in orbital mechanics. We then validated OPAS’s actual returned safe windows against this reference across five representative launch scenarios spanning low Earth orbit (LEO), a sun-synchronous orbit, and a geostationary transfer orbit.

The remainder of this report is organized as follows. Section 2 reviews relevant literature on TLE/SGP4 propagation accuracy and documented conjunction-screening practice. Section 3 describes the frozen data snapshot, reference implementation, and validation harness. Section 4 presents the audit’s findings: the original scanner’s false-safe rate, the design flaws responsible for it, an engineered fix, and a direct, case-by-case analysis of the fix’s remaining limitation. Section 5 interprets these results against documented real-world practice, Section 6 states the study’s scope boundaries, and Section 7 summarizes findings and proposes future work.

2 Background

2.1 TLE/SGP4 propagation accuracy

OPAS propagates object positions from two-line element (TLE) sets using the SGP4 model. TLE/SGP4 accuracy degrades with time since the TLE’s epoch: a commonly cited heuristic places initial error around 1 km at epoch, growing by an additional 1–3 km per day thereafter for low Earth orbit (LEO) objects [1]. An empirical comparison against a numerical integrator found SGP4-propagated position error for a real LEO object stayed within 1 km for roughly the first week from epoch, with error growing super-linearly afterward and dominated by the in-track (along-orbit) component rather than the radial or cross-track directions [2]. This distinction — error growing much faster along-track than radially — is directly relevant to how a screening radius should be shaped, discussed further in Section 5.

2.2 Documented conjunction-screening practice

NASA’s Conjunction Assessment and Collision Avoidance Best Practices Handbook states that screening volumes are assigned per protected asset and orbit regime rather than fixed [4]. One documented coarse-screening pass, used across a six-month operational dataset, applied a volume of approximately 50 km (radial) by 250 km (in-track) by 250 km (cross-track), looking several days ahead [3]. A separate documented operational example, for a near-Earth-orbit protected asset, instead uses a much tighter $0.5 \text{ km} \times 17 \text{ km} \times 20 \text{ km}$ ellipsoid, reported independently in both sources [3, 4]. A related NASA presentation states that TLE-based tracking accuracy — on the order of 1–2 km within a few days of epoch — is not considered sufficient for official conjunction assessment by the 18th Space Control Squadron, and that operators requiring rigorous collision avoidance instead submit their own ephemerides for direct screening [5].

Two points from this practice inform the present study. First, real screening volumes are anisotropic ellipsoids, not spheres: even the tightest documented example (0.5 km radial) uses a substantially larger in-track and cross-track extent (17–20 km), reflecting the in-track-dominated error behavior discussed above. Second, screening volumes are assigned per mission rather than fixed, and even the largest documented example (50 km radial) was used only as an initial coarse filter within one study, not applied universally as a final safety threshold — a distinction relevant to evaluating OPAS’s single, final, universally-applied 50 km spherical radius (Section 4).

3 Methodology

3.1 Frozen data snapshot

To ensure every experiment in this study ran against identical data, the live `opas_db.debris` MongoDB collection was exported once to a static, checksummed snapshot (32,517 objects, SHA-256 verified before every subsequent script run). TLE age was computed relative to this snapshot’s export time throughout, rather than the wall clock, so that results are reproducible independent of when the analysis scripts are actually executed. All measurements below ran against this single snapshot and the unmodified OPAS source at git commit `a89f2e72`².

3.2 Baseline measurement

The unmodified `scan_windows` function was run against five launch presets spanning a range of orbital regimes: an ISS resupply orbit (420 km, 51.6° inclination), a Starlink deployment orbit (550 km, 53.0°), a sun-synchronous orbit (705 km, 98.2°), a polar orbit (800 km, 90°), and a geostationary transfer orbit (35,786 km, 6.0°), each scanned over a 6-hour horizon. This produced, for each preset, the exact set of “safe windows” OPAS itself would report to a user.

3.3 Dense reference implementation

To evaluate whether those reported windows were actually free of close approaches, we built an independent reference calculation using the same orbital data, propagation model (SGP4 via Skyfield), and simplified circular-orbit vehicle trajectory as OPAS, differing only in sampling density and closest-approach interpolation: object and vehicle positions were sampled at approximately 1-second resolution (versus OPAS’s coarser internal sampling, detailed in Section 4), and the true closest-approach distance between sample points was recovered via parabolic interpolation of the three nearest samples — exact for locally straight-line relative motion. Because both OPAS and the reference share the same underlying orbital mechanics, any disagreement between them is attributable to how finely closest approach is resolved rather than a difference in physics.

The reference implementation was validated before use: a self-test using two planted objects with known closest-approach geometry recovered the correct minimum distance to within 0.02 km after parabolic refinement, and an independent comparison of 1-second versus 2-second sampling across 38,000 object-launch pairs showed a median disagreement of 0.5 m with zero classification flips at any of four candidate screening radii (5, 10, 25, and 50 km), confirming that 1-second sampling is dense enough to serve as ground truth at the distances relevant to this study.

3.4 Validating OPAS’s reported windows

For each preset, every window OPAS’s baseline run reported as safe was checked minute-by-minute against the dense reference at OPAS’s own per-object screening radius, yielding a *false-safe rate*: the fraction of a reported window’s duration during which the reference showed at least one tracked object inside that same radius. A parameter sweep across OPAS’s internal sampling settings (coarse step, fine step, and inner-orbit sample count) was additionally run to test whether the false-safe rate was attributable to sampling density specifically, as opposed to some other property of the scanning algorithm. The parameterized scanner used for this sweep was confirmed to reproduce `scan_windows`’s baseline output exactly (identical windows and evaluation counts) for the three presets it was run against (ISS, Starlink, SSO) before its results were used.

²Python 3.13.14, NumPy 2.4.4, Skyfield 1.54; OPAS’s optional native C++ distance-check extension was active for all runs.

3.5 Engineered fix and validation

Based on the findings in Section 4, a corrected scanner was implemented with four changes: (i) trajectory sampling density increased from roughly a dozen to 600 points per orbit, (ii) the per-object screening radius rescaled from a flat 50 km with an exponential TLE-age penalty to a 10 km base with a capped linear age penalty (Section 5), (iii) temporal verification tightened from 2-minute to 1-minute resolution, and (iv) returned windows clamped to the requested scan range. This corrected scanner’s output was validated against the dense reference using the exact per-object radius formula it applies internally — rather than an approximate flat radius — so that the reported false-safe rate reflects the scanner’s own actual decision criterion precisely. Where a non-zero false-safe rate remained, its cause was examined directly: for each reference-confirmed close approach inside a corrected-scanner “safe” window, the scanner’s own sampled positions at that exact launch time were recomputed to confirm whether its own math, run on its own inputs, indeed reported no threat. Separately, every contiguous stretch of reference-confirmed clear one-minute launch times was enumerated directly from the reference data, independent of the scanner’s own window-forming algorithm, to test whether a genuinely clear window of useful length exists regardless of whether the scanner itself located it.

4 Results

4.1 Baseline: reported safe windows were false-safe in crowded shells

Table 1 summarizes every window OPAS’s unmodified `scan_windows` reported across the five presets, and what fraction of each window’s duration the dense reference showed was actually within OPAS’s own screening radius of a tracked object.

Preset	Window duration	Reference samples	False-safe rate
ISS (420 km)	24 min	24	100%
Starlink (550 km)	46 min	46	100%
Starlink (550 km)	16 min	16	100%
SSO (705 km)	20 min	20	100%
GEO transfer	360 min (clipped)	360	0%

Table 1: Baseline false-safe rate for every window OPAS reported as safe, across a 6-hour scan horizon. Polar orbit reported no windows and is omitted.

Every window reported in the three crowded low-Earth-orbit shells (ISS, Starlink, SSO) was, per the dense reference, within OPAS’s own danger radius for the entirety of its reported duration. GEO is the control case: with only 840 tracked objects spread across a far larger volume, OPAS’s claim of safety there was correct.

4.2 Root cause: coarse sampling, not incorrect physics

Three properties of the unmodified scanner jointly explain Table 1. First, `scan_windows`’ inner check samples each orbit at only 12 intervals (about 14 points, roughly one every 6 minutes), against a 50 km screening radius; the dense reference showed that at this radius, 100% of one-minute launch times across the scan horizon had at least one object within range for ISS, Starlink, and SSO alike (mean 8.8–12.2 objects per launch), meaning a real encounter has only a small chance of landing on one of the sampled points. Second, a parameter sweep across the coarse step, fine step, and inner-orbit sample count is consistent with this: doubling the inner sample count from 12 to 24 reduced the reported window count to zero for all three crowded shells at every coarse step tested, rather than producing narrower but genuinely safe windows.

Third, an independent boundary-handling defect was identified: the refinement loop that trims a candidate window’s start and end does not clamp to the originally requested scan range, occasionally extending a reported window several minutes beyond what was actually evaluated.

4.3 Engineered fix and its residual limitation

A corrected scanner was implemented with four changes (Section 3): 600-point trajectory sampling (versus roughly a dozen points), a screening radius of 10–15 km scaled by TLE age (versus a flat 50 km with an uncapped exponential age penalty reaching up to 200 km), 1-minute temporal verification (versus 2-minute), and clamping of returned windows to the requested horizon. Applied to the ISS and Starlink presets, this replaced each preset’s single, entirely false-safe window with one new window per preset (Table 2); SSO returned no windows, addressed directly below.

Preset	Window	Reference samples	Marked unsafe	False-safe rate
ISS	10:33–11:25 (52 min)	52	5	9.6%
Starlink	10:03–10:22 (19 min)	19	4	21.1%

Table 2: Corrected scanner’s reported windows, validated at the scanner’s exact per-object radius formula (10–15 km by TLE age). SSO returned no window.

The residual false-safe rate in both windows was examined case by case. For each of the 10 reference-confirmed object–launch-time pairs within these two windows’ combined 9 unsafe launch minutes (5 for ISS, 4 for Starlink), the corrected scanner’s own 600-point sampled positions were recomputed at the exact launch instant the reference flagged. All ten (100%) were confirmed to be genuinely invisible to the scanner’s own math: its sampled minimum distance at that launch time remained well outside its radius even though the reference’s interpolation-refined minimum was inside it — consistent with the scanner’s ~ 9.3 – 9.6 -second waypoint spacing (which varies slightly with each preset’s orbital period) versus LEO relative velocities of up to ~ 15 km/s, which can move two objects up to ~ 135 – 145 km between consecutive samples.

4.4 Direct verification of clear-window existence

The corrected scanner’s coarse-to-refined algorithm examines at most five candidate coarse-safe runs per scan (Section 3), and, as shown above, a window it does report is not guaranteed to be genuinely clear. To test both possibilities independently of the scanner’s own algorithm, every contiguous stretch of reference-confirmed clear one-minute launch times was directly enumerated across the full 6-hour horizon at the corrected radius, for all three crowded shells.

No stretch of at least 15 minutes exists for Starlink (longest 14 min) or SSO (longest 6 min): no interval of the reported Starlink window’s own length (19 min) is genuinely clear anywhere in the horizon, consistent with its 21.1% false-safe measurement in Table 2, and SSO’s absence of any reported window is correct. ISS, by contrast, contains five genuinely clear stretches of 15–19 minutes across the horizon; only part of one overlaps the corrected scanner’s single reported window, which is itself only partially clear (Table 2). The corrected scanner’s five-run cap is therefore not guaranteed to locate every genuine clear interval that exists, independent of its residual false-safe rate.

5 Discussion

5.1 The original 50 km radius was never meant to be a final threshold

OPAS’s original screening radius (50 km, before TLE-age inflation) is coincidentally close in magnitude to the 50 km radial extent of one documented NASA coarse-screening volume [3]. The critical difference is that NASA’s screening volumes are explicitly assigned per protected asset and orbit regime rather than fixed, and even this large example was used only as an initial filter within a six-month study, not applied universally as a final safety threshold; a separate documented operational example for a near-Earth-orbit asset uses a radius of just 0.5 km [3, 4]. OPAS instead applies a single 50 km radius universally, as the sole and final safety criterion, regardless of orbit regime or mission. This reframes Section 4’s finding: the radius itself was not unreasonable as one possible coarse filter, but using it as a final, one-size-fits-all decision threshold — combined with under-sampling that never properly evaluated even that generous radius — produced a tool that reported safety where none existed.

5.2 The corrected radius is an assumption, not literature-derived

The corrected scanner’s screening radius (a 10 km base plus 0.5 km per day of TLE age, capped at 5 km) was chosen for tractability, not derived from measured SGP4 error rates or from the screening volumes in Section 2. Because the median TLE age in the snapshot is under one day, the operative radius for most objects is close to the 10 km base. Its relationship to the literature is not one-directional: the 1–3 km/day heuristic [1] implies a steeper age penalty than the 0.5 km/day used here, whereas the single-object measurement in [2] and the 0.5 km radial extent of a documented NASA operational volume [3, 4] suggest a 10 km sphere is generous in the radial direction. The Starlink and SSO result is correspondingly radius-sensitive: in this study’s reference data, the fraction of launch times with at least one object inside a flat radius rises from about 10–11% at 5 km to 29–45% at 10 km for those two shells. We therefore present the results in Section 4 as a property of the corrected radius, not of any documented operational threshold, whereas the baseline finding, measured at OPAS’s own radius, does not depend on it.

5.3 An isotropic radius is itself a simplification

Both OPAS’s original scanner and the corrected version in this study use a single spherical radius around each tracked object. Documented practice uses an anisotropic ellipsoid instead — far tighter in the radial direction than along-track or cross-track [3, 4] — because, as shown in Section 2, SGP4/TLE position error is dominated by in-track drift rather than radial displacement. A sphere large enough to cover realistic in-track error will therefore be much larger than necessary in the radial direction, over-restricting some genuinely safe launch times, while a sphere sized to radial error would under-cover the in-track direction. This is a structural limitation of OPAS’s design, not something the parameter changes made in this study could resolve, and is proposed as future work in Section 7.

5.4 The residual false-safe rate has one confirmed cause

Section 4 traced the corrected scanner’s remaining 9.6–21.1% false-safe rate, in both windows it reported, entirely to finite waypoint spacing: every one of the ten reference-confirmed close approaches examined was invisible to the scanner’s own sampled positions at the exact launch instant the reference flagged, consistent with ~ 9.3 – 9.6 -second sample spacing against LEO relative velocities of up to ~ 15 km/s. No fixed sampling interval can guarantee catching every possible close approach at these velocities without approaching the computational cost of continuous collision detection.

6 Limitations

Scope. This study evaluates only OPAS’s `/safe-windows` screening-radius logic (`scan_windows` and its inner threat check). OPAS’s separate probability-of-collision scoring, used by its `/alert` endpoint, was not evaluated and is outside this study’s scope.

Trajectory model. Both OPAS and this study’s reference implementation model the launch vehicle’s path as a simplified circular orbit derived from launch site, target altitude, and inclination. Real launch trajectories — particularly the ascent phase, deliberately excluded from both OPAS’s inner check and this study’s reference to keep the comparison like-for-like — follow a more complex path shaped by vehicle performance and guidance constraints. Findings in this report characterize OPAS’s own model, not real-world launch trajectories.

Single data snapshot. All measurements ran against one frozen catalog snapshot (Section 3). Debris populations and TLE freshness vary over time; the specific false-safe rates reported here are a measurement at one point in time, not a permanent property of OPAS independent of catalog state. The qualitative finding — that coarse sampling can hide close approaches regardless of catalog density — is expected to generalize, but the specific percentages in Table 1 and Table 2 should not be read as universal constants.

Screening radius. As discussed in Section 5, the corrected scanner’s 10 km base radius and TLE-age scaling are engineering choices, not values derived from measured propagation error, and the Starlink and SSO results are sensitive to them.

Candidate-run cap. `scan_windows`, and the corrected scanner which inherits its structure, examines at most five raw coarse-safe candidate runs per scan. Section 4 shows this can cause a genuinely clear window to go unreported even where one exists elsewhere in the scanned horizon, independent of the false-safe rate discussed above.

Reference implementation precision. The dense reference implementation’s launch-time grid is built from an actual floating-point sample spacing rather than exact clock time, and could in principle drift from true clock time over a multi-hour scan; no such drift was found to affect any of the ten residual cases examined in Section 4, though this was not independently checked across the reference’s full coverage. Confidence in the reference as ground truth is otherwise supported by its independent validation (Section 3).

7 Conclusion and Future Work

This study audited OPAS’s `/safe-windows` scanner against an independently-implemented, validated dense reference, and found that every window reported as safe in three of five tested orbital regimes was, in fact, unsafe for its entire reported duration. This was traced to insufficient trajectory sampling relative to the object density in crowded low-Earth-orbit shells, not to an error in the underlying orbital mechanics. A corrected scanner — denser sampling, a rescaled screening radius, and finer temporal verification — eliminated the false-safe problem’s dominant cause. Applied to the three crowded shells, it returned one window each for ISS and Starlink, with a residual false-safe rate (9.6% and 21.1% respectively) traced entirely to finite waypoint spacing, and no window for SSO. Direct enumeration of clear intervals in the dense reference, independent of the scanner’s own algorithm, confirmed that Starlink and SSO genuinely have no clear stretch of at least 15 minutes at the corrected radius — so the reported Starlink window is itself false-safe — while also showing that the scanner’s five-candidate-run cap can cause it to miss genuinely clear ISS windows that do exist.

Three directions follow directly from the limitations discussed in Section 6. First, replacing the isotropic screening radius with an anisotropic ellipsoid — tighter radially than along-track, consistent with documented SGP4 error behavior and real conjunction-screening practice — would likely change which launch times are flagged unsafe. Second, tying the TLE-age radius penalty to measured propagation-error rates, rather than the tractability-driven formula used

here, would ground the corrected scanner’s radius in evidence; its effect on the number of safe windows cannot be predicted from this study’s data alone. Third, the 9-second waypoint gap identified as the corrected scanner’s sole confirmed residual failure mode is a fundamental limit of discrete point-sampling; closing it fully would require either continuous swept-volume collision detection or a sampling density approaching the reference implementation’s own computational cost — a genuine engineering trade-off between accuracy and the response-time constraints of a live web API, rather than a simple parameter change.

References

- [1] C. Priestley and W. Handley, “jaxsgp4: GPU-accelerated mega-constellation propagation with batch parallelism,” *arXiv preprint arXiv:2603.27830*, 2026.
- [2] A. T. Rich, “Investigating Analytical and Numerical Methods to Predict Satellite Orbits Using Two-Line Element Sets,” thesis, Air Force Institute of Technology, 2017.
- [3] M. D. Hejduk and D. A. Pachura, “Conjunction assessment screening volume sizing and event filtering in light of natural conjunction event development behaviors,” *AAS/AIAA Astrodynamics Specialist Conference*, Stevenson, WA, Aug. 2017.
- [4] NASA Office of the Chief Engineer, “NASA Spacecraft Conjunction Assessment and Collision Avoidance Best Practices Handbook,” NASA/SP-20230002470, Rev. 1, Feb. 2023.
- [5] A. Mashiku, “NASA spacecraft conjunction assessment and collision avoidance best practices: The history, evolution and opportunities,” presentation, *44th COSPAR Scientific Assembly*, Athens, Greece, July 2022.